TASK-3

```sql
CREATE TABLE sales (
    Order_ID INTEGER PRIMARY KEY,
    Order_Date TEXT,
    Product TEXT,
    Category TEXT,
    Quantity REAL,
    Unit_Price REAL,
    Sales_Amount INTEGER,
    Region TEXT,
    Quantity_Flag TEXT
);
```

```sql
INSERT INTO sales
(Order_ID, Order_Date, Product, Category, Quantity,
 Unit_Price, Sales_Amount, Region, Quantity_Flag)
VALUES
(1001, '2026-01-05', 'Laptop', 'Electronics', 2, 50000, 100000, 'South', 'Valid');

INSERT INTO sales VALUES
(1002, '2026-01-06', 'Smartphone', 'Electronics', 3, 20000, 60000, 'North', 'Valid');

INSERT INTO sales VALUES
(1003, '2026-01-07', 'Office Chair', 'Furniture', 5, 3000, 15000, 'East', 'Valid');

INSERT INTO sales VALUES
(1004, '2026-01-08', 'Headphones', 'Electronics', 4, 1500, 6000, 'West', 'Valid');

INSERT INTO sales VALUES
(1005, '2026-01-09', 'Desk', 'Furniture', 2, 7000, 14000, 'South', 'Valid');

INSERT INTO sales VALUES
(1007, '2026-01-11', 'Laptop', 'Electronics', 1000, 50000, 50000000, 'East', 'Valid');
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region	Quantity_Flag
1	1001	2026-01-05	Laptop	Electronics	2.0	50000.0	100000	South	Valid
2	1002	2026-01-06	Smartphone	Electronics	3.0	20000.0	60000	North	Valid
3	1003	2026-01-07	Office Chair	Furniture	5.0	3000.0	15000	East	Valid
4	1004	2026-01-08	Headphones	Electronics	4.0	1500.0	6000	West	Valid
5	1005	2026-01-09	Desk	Furniture	2.0	7000.0	14000	South	Valid
6	1007	2026-01-11	Laptop	Electronics	1000.0	50000.0	50000000	East	Valid
7	1008	2026-01-12	Smartphone	Electronics	2.0	20000.0	40000	South	Valid
8	1009	2026-01-13	Office Chair	Furniture	3.0	3000.0	9000	West	Valid
9	1012	2026-01-16	Keyboard	Electronics	5.0	800.0	3500	East	Valid

Query 1: Display all records

```sql
SELECT *
FROM sales
ORDER BY Order_ID;
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region	Quantity_Flag
1	1001	2026-01-05	Laptop	Electronics	2.0	50000.0	100000	South	Valid
2	1002	2026-01-06	Smartphone	Electronics	3.0	20000.0	60000	North	Valid
3	1003	2026-01-07	Office Chair	Furniture	5.0	3000.0	15000	East	Valid
4	1004	2026-01-08	Headphones	Electronics	4.0	1500.0	6000	West	Valid
5	1005	2026-01-09	Desk	Furniture	2.0	7000.0	14000	South	Valid
6	1007	2026-01-11	Laptop	Electronics	1000.0	50000.0	50000000	East	Valid
7	1008	2026-01-12	Smartphone	Electronics	2.0	20000.0	40000	South	Valid
8	1009	2026-01-13	Office Chair	Furniture	3.0	3000.0	9000	West	Valid
9	1012	2026-01-16	Keyboard	Electronics	5.0	800.0	3500	East	Valid

Query 2: Find high-value orders

```sql
SELECT Order_ID, Order_Date, Product, Sales_Amount, Region
FROM sales
WHERE Sales_Amount > 10000
ORDER BY Sales_Amount DESC;
```

	Order_ID	Order_Date	Product	Sales_Amount	Region
1	1007	2026-01-11	Laptop	50000000	East
2	1001	2026-01-05	Laptop	100000	South
3	1002	2026-01-06	Smartphone	60000	North
4	1008	2026-01-12	Smartphone	40000	South
5	1003	2026-01-07	Office Chair	15000	East
6	1005	2026-01-09	Desk	14000	South

Query 3: Sort sales from highest to lowest

```sql
SELECT Order_ID, Product, Sales_Amount
FROM sales
ORDER BY Sales_Amount DESC;
```

	Order_ID	Product	Sales_Amount
1	1007	Laptop	50000000
2	1001	Laptop	100000
3	1002	Smartphone	60000
4	1008	Smartphone	40000
5	1003	Office Chair	15000
6	1005	Desk	14000
7	1009	Office Chair	9000
8	1004	Headphones	6000
9	1012	Keyboard	3500

Query 4: Calculate total sales by region

```sql
SELECT
    Region,
    COUNT(*) AS Order_Count,
    SUM(Sales_Amount) AS Total_Sales
FROM sales
GROUP BY Region
ORDER BY Total_Sales DESC;
```

	Region	Order_Count	Total_Sales
1	East	3	50018500
2	South	3	154000
3	North	1	60000
4	West	2	15000

Query 5: Calculate average sales by category

```
68  SELECT
69      Category,
70      ROUND(AVG(Sales_Amount), 2) AS Average_Sales,
71      SUM(Quantity) AS Total_Quantity
72  FROM sales
73  GROUP BY Category
74  ORDER BY Average_Sales DESC;
75
76
77
78
79
```

	Category	Average_Sales	Total_Quantity
1	Electronics	8368250.0	1016.0
2	Furniture	12666.67	10.0

Query 6: Search for products containing "phone"

```
76  SELECT Order_ID, Product, Category, Sales_Amount
77  FROM sales
78  WHERE Product LIKE '%phone%'
79  ORDER BY Product;
80
81
```

	Order_ID	Product	Category	Sales_Amount
1	1004	Headphones	Electronics	6000
2	1002	Smartphone	Electronics	60000
3	1008	Smartphone	Electronics	40000

Query 7: Calculate sales within a date range

```sql
SELECT
    COUNT(*) AS Order_Count,
    SUM(Sales_Amount) AS Total_Sales,
    ROUND(AVG(Sales_Amount), 2) AS Average_Sale
FROM sales
WHERE Order_Date BETWEEN '2026-01-01' AND '2026-01-15';
```

	Order_Count	Total_Sales	Average_Sale
1	8	50244000	6280500.0